椭圆的定义与基本要素

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

$$A.\frac{x^2}{45}+\frac{y^2}{36}=1$$
$$B.\frac{x^2}{36}+\frac{y^2}{27}=1$$
$$C.\frac{x^2}{27}+\frac{y^2}{18}=1$$
$$D.\frac{x^2}{18}+\frac{y^2}{9}=1$$

椭圆图形化定义

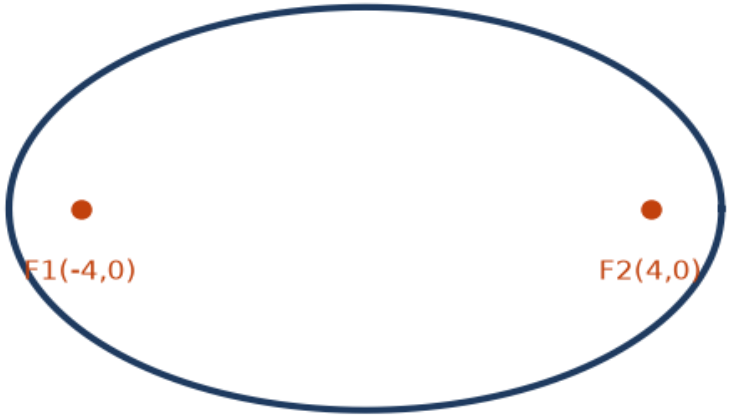
$$E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b > 0)$$

$A. \frac{x^2}{45} + \frac{y^2}{36} = 1$

$B. \frac{x^2}{36} + \frac{y^2}{27} = 1$

$C. \frac{x^2}{27} + \frac{y^2}{18} = 1$

$D. \frac{x^2}{18} + \frac{y^2}{9} = 1$



椭圆标准方程的推导

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

$$A.\frac{x^2}{45}+\frac{y^2}{36}=1$$
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椭圆标准方程公式聚焦

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

$$A.\frac{x^2}{45}+\frac{y^2}{36}=1$$
$$B.\frac{x^2}{36}+\frac{y^2}{27}=1$$
$$C.\frac{x^2}{27}+\frac{y^2}{18}=1$$

例题精讲：基础例题

$$A(x_1, y_1), B(x_2, y_2)$$

$(5 分) 已知椭圆 E: \frac{x^{2}}{a^{2}}+\frac

易错辨析：椭圆标准方程的常见错误

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

$$A.\frac{x^2}{45}+\frac{y^2}{36}=1$$
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$$C.\frac{x^2}{27}+\frac{y^2}{18}=1$$
$$D.\frac{x^2}{18}+\frac{y^2}{9}=1$$

对比辨析：椭圆标准方程的焦点位置

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

$$A.\frac{x^2}{45}+\frac{y^2}{36}=1$$
$$B.\frac{x^2}{36}+\frac{y^2}{27}=1$$
$$C.\frac{x^2}{27}+\frac{y^2}{18}=1$$
$$D.\frac{x^2}{18}+\frac{y^2}{9}=1$$

课堂练习：基础练习

$$\frac{x^2}{12}+\frac{y^2}{3}=1$$

$(5 分) 已知双曲线 C: \frac{x^{2}}{a^{2}}-\fra

例题精讲：变式例题

$$\because y=\frac{x}{x+2}$$

$(5 分) 曲线 y=\frac{x}{x+2} 在点 (-1,-1) 处的切

课堂练习：巩固练习

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(5 分) 已知双曲线 E 的中心为原点, P(3,0) 是 E 的焦点, 过

课堂小结：椭圆及其标准方程

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

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$$B.\frac{x^2}{36}+\frac{y^2}{27}=1$$
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公式聚焦：椭圆标准方程的几何意义

$$E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$$

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$$C.\frac{x^2}{27}+\frac{y^2}{18}=1$$